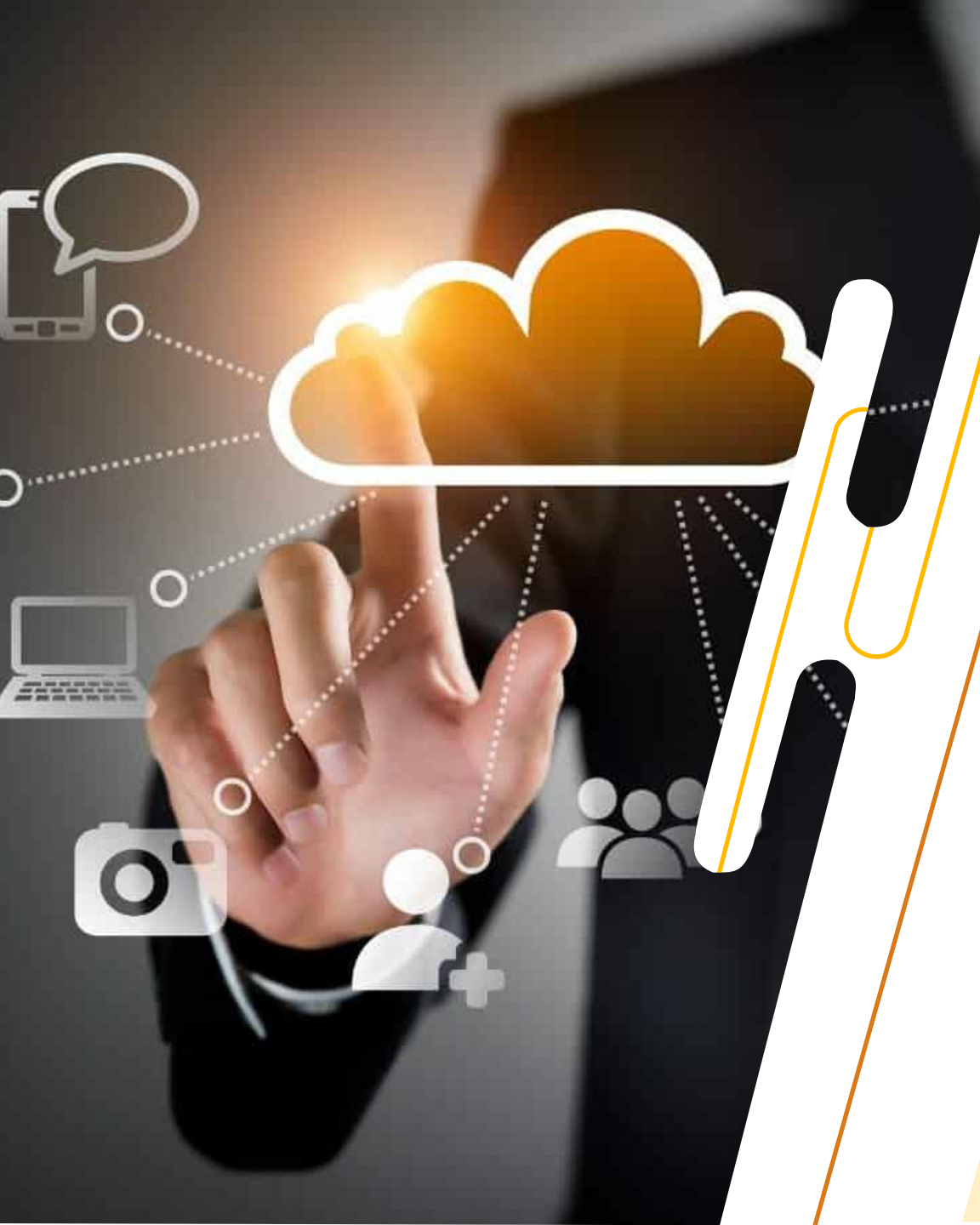




IT3061

Massive Data Processing and Cloud Computing

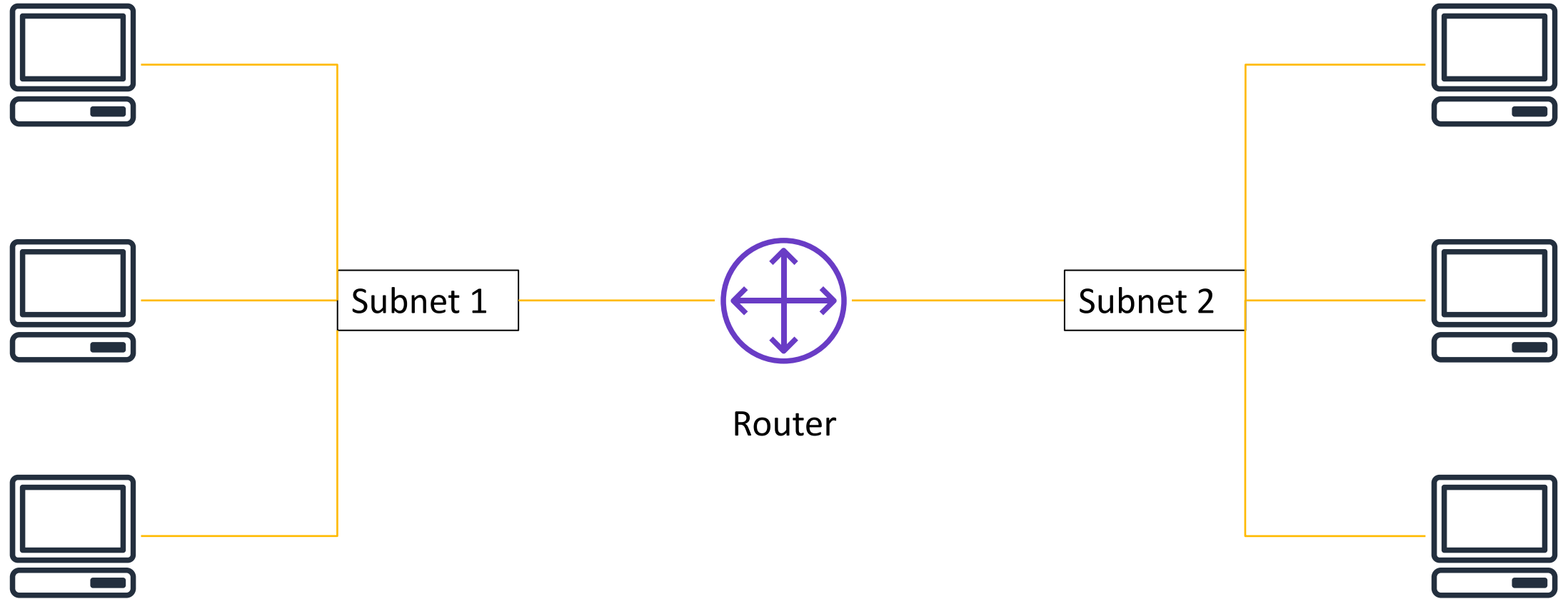
YEAR 3 SEMESTER 2
DATA SCIENCE SPECIALIZATION

Lecture Outline

- 1. Networking Basics
- 2. Cloud Networking
- 3. Benefits of Cloud Networking
- 4. Virtual Networks
- 5. Subnets
- 6. Access Controlling
- 7. Network Gateway Services
- 8. Network Peering

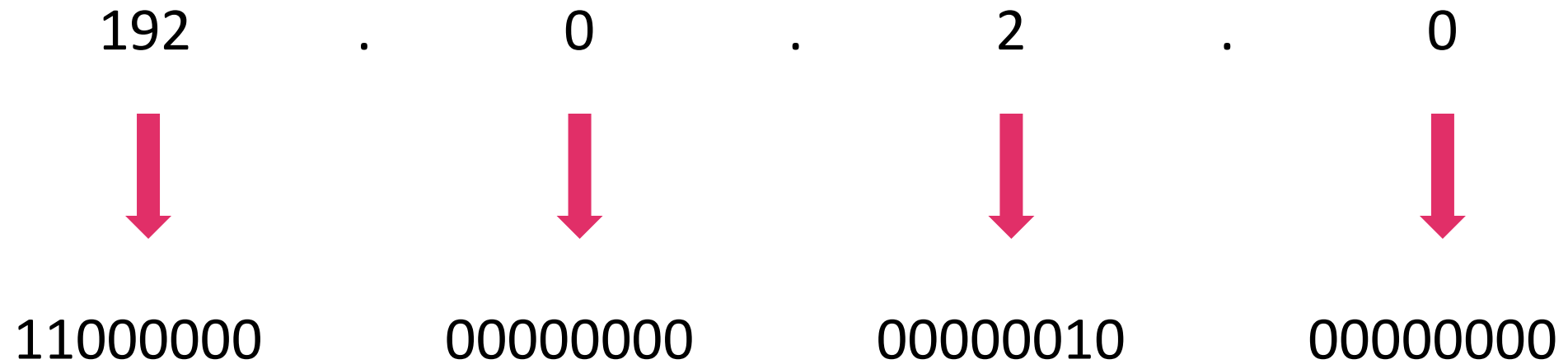


Networking Basics



Networking Basics

- IP Addresses



Networking Basics

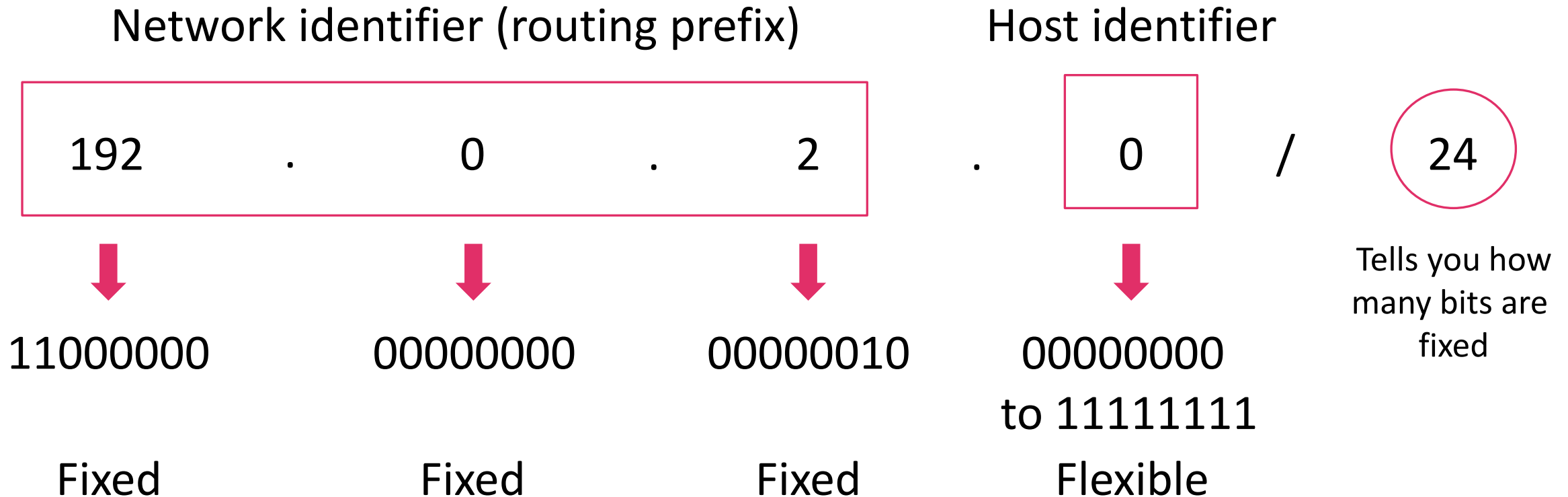
- IPv4 and IPv6 Addresses

IPv4 (32-bit) address: 192.0.2.0

IPv6 (128-bit) address: 2600:1f18:22ba:8c00:ba86:a05e:a5ba:00FF

Networking Basics

- Classless Inter-Domain Routing (CIDR)



Networking Basics

- **Open Systems Interconnection (OSI) model**

Layer	Number	Function	Protocol/ Address
Application	7	Means for an application to access a computer network	HTTP(S), FTP, DHCP, LDAP
Presentation	6	• Ensures that the application layer can read the data • Encryption	ASCII, ICA
Session	5	Enables orderly exchange of data	NetBIOS, RPC
Transport	4	Provides protocols to support host-to-host communication	TCP, UDP
Network	3	Routing and packet forwarding (routers)	IP
Data link	2	Transfer data in the same LAN network (hubs and switches)	MAC
Physical	1	Transmission and reception of raw bitstreams over a physical medium	Signals (1s and 0s)

Cloud Networking

- Cloud Networking is the use of cloud-based services to deploy a corporate network that connects an organization's employees, resources, and applications.
- Traditionally, organizations used their own private network hardware components to create an isolated wide area network (WAN) for secure communication and application deployment. Setting up and managing the network was expensive and complicated.
- Cloud networking solves these challenges by allowing organizations to use virtual network components instead.

Benefits of Cloud Networking

- Efficient network management
- Increased scalability
- Easier security
- Improved monitoring and maintenance

Virtual Networks

- Logically isolated from other virtual networks.
- Has one or more IP ranges.
- Has one or more subnets.
- Can peer with other virtual networks if IP ranges do not overlap.
- Examples :-

AWS VPC (Virtual Private Cloud)

Azure Virtual Network

Subnets

- Segments virtual network IP range even further
- Subnet CIDR blocks cannot overlap within a virtual network
- Traffic can be restricted using Access Control Lists
- AWS offers private and public subnets

Instances in public subnets can send outbound traffic directly to internet.

Instances in private subnets cannot send outbound traffic directly to internet.

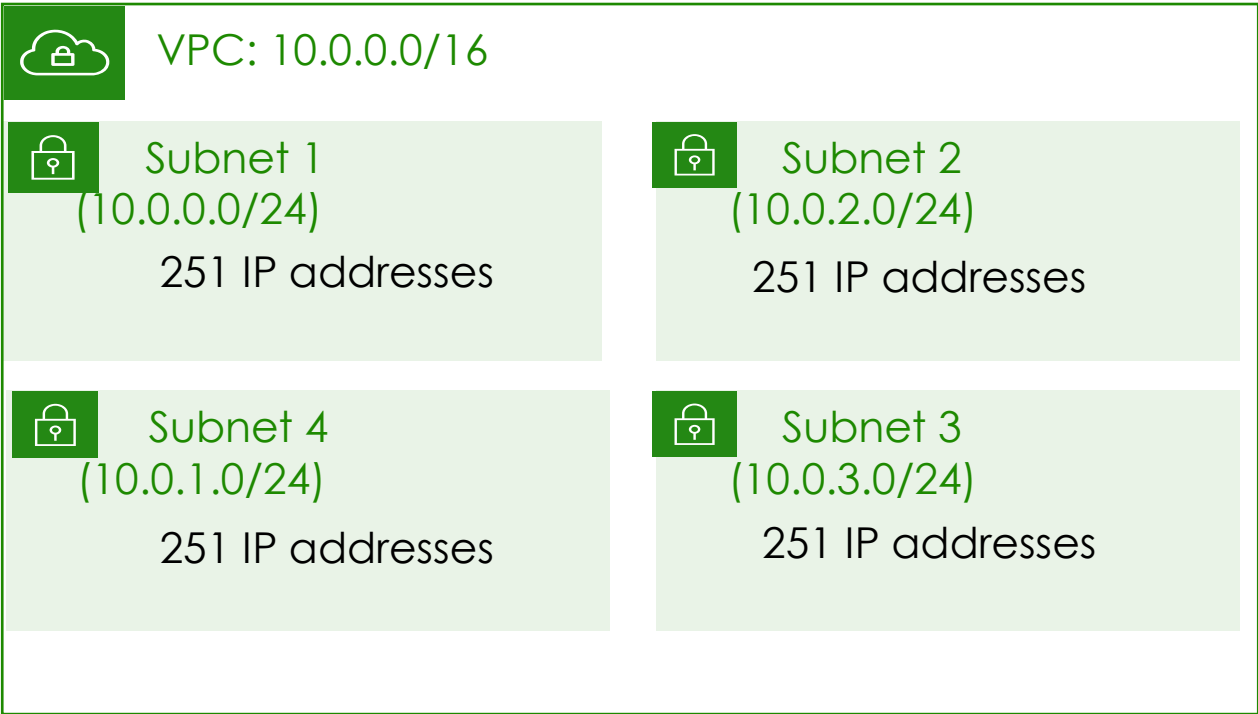
- Azure does not have the public/private subnet concept.

All subnets can be considered as public subnets

Can implement private subnet functionality through route tables

Reserved IP Addresses

Example: A VPC with an IPv4 CIDR block of 10.0.0.0/16 has 65,536 total IP addresses. The VPC has four equal-sized subnets. Only 251 IP addresses are available for use by each subnet.



IP Addresses for CIDR block 10.0.0.0/24	Reserved for
10.0.0.0	Network address
10.0.0.1	Internal communication
10.0.0.2	Domain Name System (DNS) resolution
10.0.0.3	Future use
10.0.0.255	Network broadcast address

Reserved IP Addresses

- In both AWS and Azure reserves the first four addresses and the last address, for a total of five IP addresses within each subnet.

- Example : 10.0.0.0/24

1st IP - 10.0.0.0 – Network Address

2nd IP - 10.0.0.1 – Reserved

3rd IP - 10.0.0.2 – Reserved

4th IP - 10.0.0.3 – Reserved

Last IP - 10.0.0.255 – Broadcast Address

First usable IP - 10.0.0.4

Last usable IP - 10.0.0.254

Number of usable IPs - $28 - 5 = 251$

Access Controlling

- Implements a simple firewall functionality to control inbound and outbound traffic to a subnet / virtual network interface.
- Need to add Access Control Lists (ACL) rules which contains
 - Rule number
 - Protocol
 - Source / Destination Port Ranges
 - Source / Destination IP Ranges
 - Allow / Deny Action
- Usage varies in CSPs
 - AWS Security Groups – Assigned to an EC2 instance
 - AWS Network Access Control List (NACL) – Assigned to a subnet
 - Azure Network Security Groups (NSG) – Can be assigned to both subnets and NICs

Route Tables

- A route table contains a set of rules, called routes, that specifies how packets should be routed in a virtual network.
- Route tables are associated to subnets, and each packet leaving a subnet is handled based on the associated route table.
- Each route table can be associated to multiple subnets, but a subnet can only be associated to a single route table.

Network Gateway Services

Virtual Network Functions associated with providing connectivity between cloud resources and outside networks / internet.

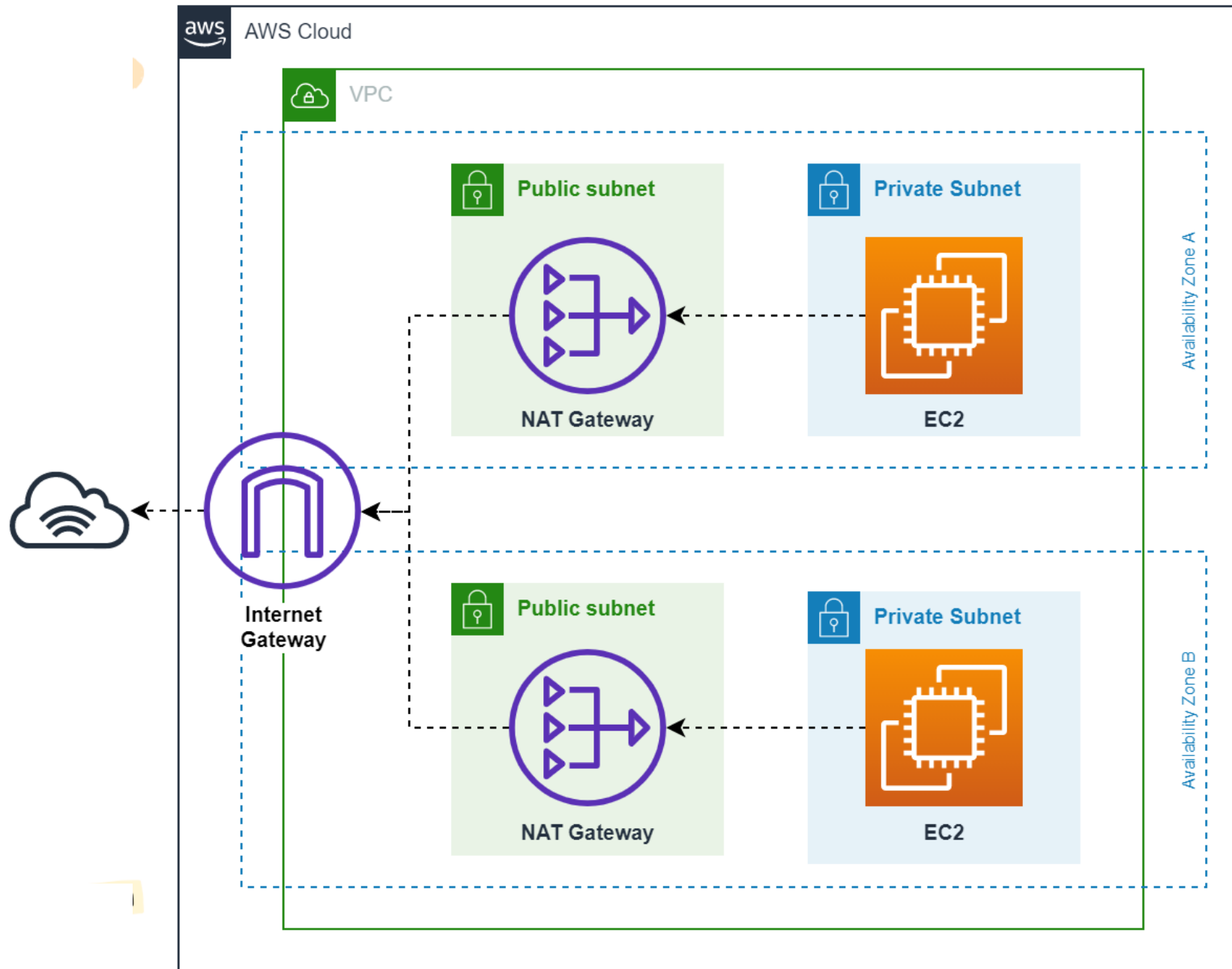
- Internet Gateway
- NAT Gateway
- VPN Gateway
- Network Transit Gateway

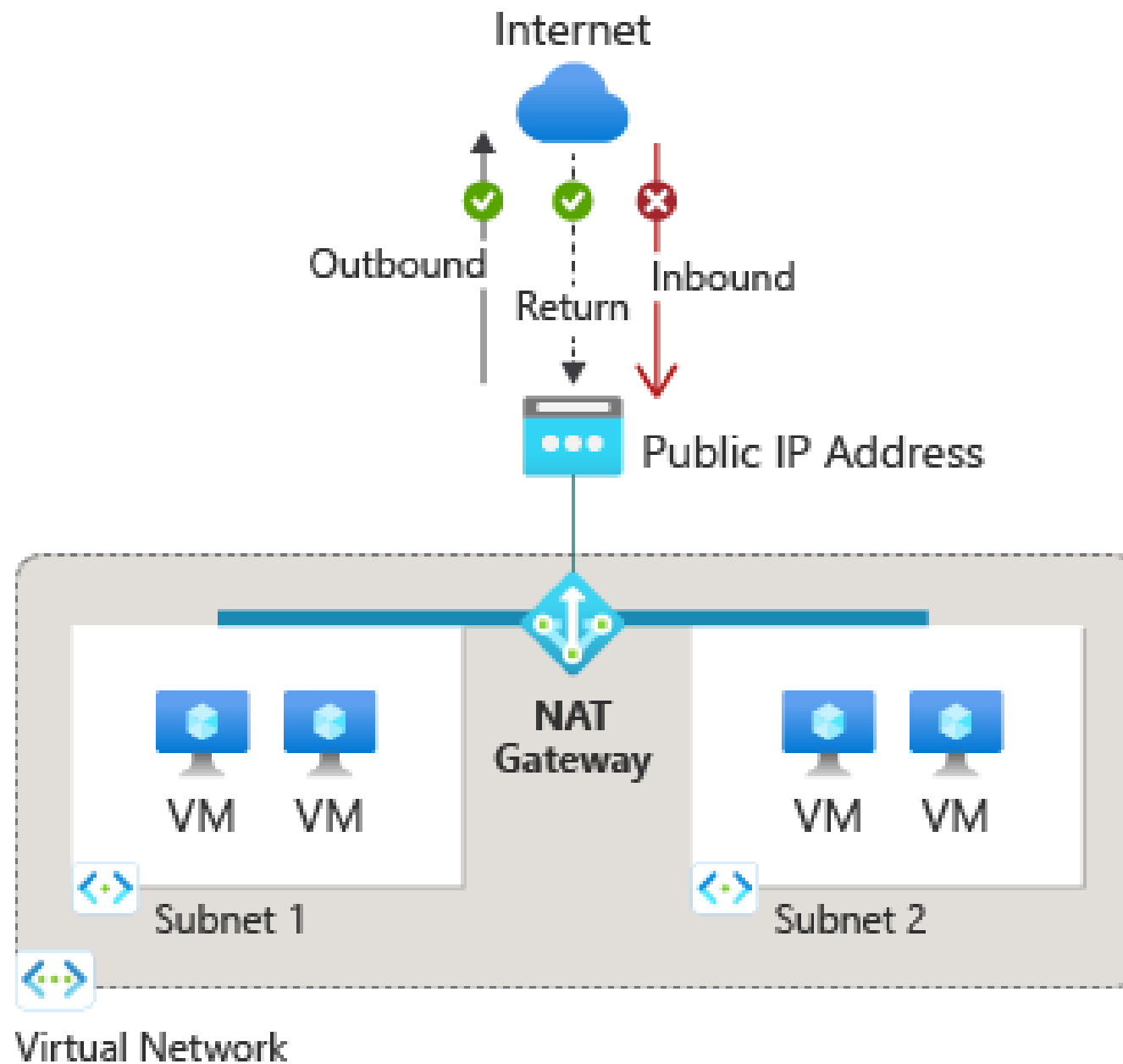
Internet Gateway

- Provides connectivity between internet and virtual network
- Scales horizontally
- No bandwidth constraints
- Implementations
 - AWS Internet Gateway
 - Azure does not have the internet gateway concept. Internet connectivity is provided by default through the default gateway

NAT Gateway

- Provides Network Address Translation (NAT) to compute instances in private subnets
- Prevents connections to subnets getting initiated from the internet.
- NAT Gateway should be deployed in a public subnet.
- Implementations
 - AWS NAT Instance
 - AWS NAT Gateway
 - Azure Virtual Network NAT





VPN Gateway

- VPN Gateways are used to create encrypted tunnels over the public internet to send encrypted traffic between virtual networks and on-premises networks.
- Implementations
 - Azure VPN Gateway
 - AWS Virtual Private Gateway

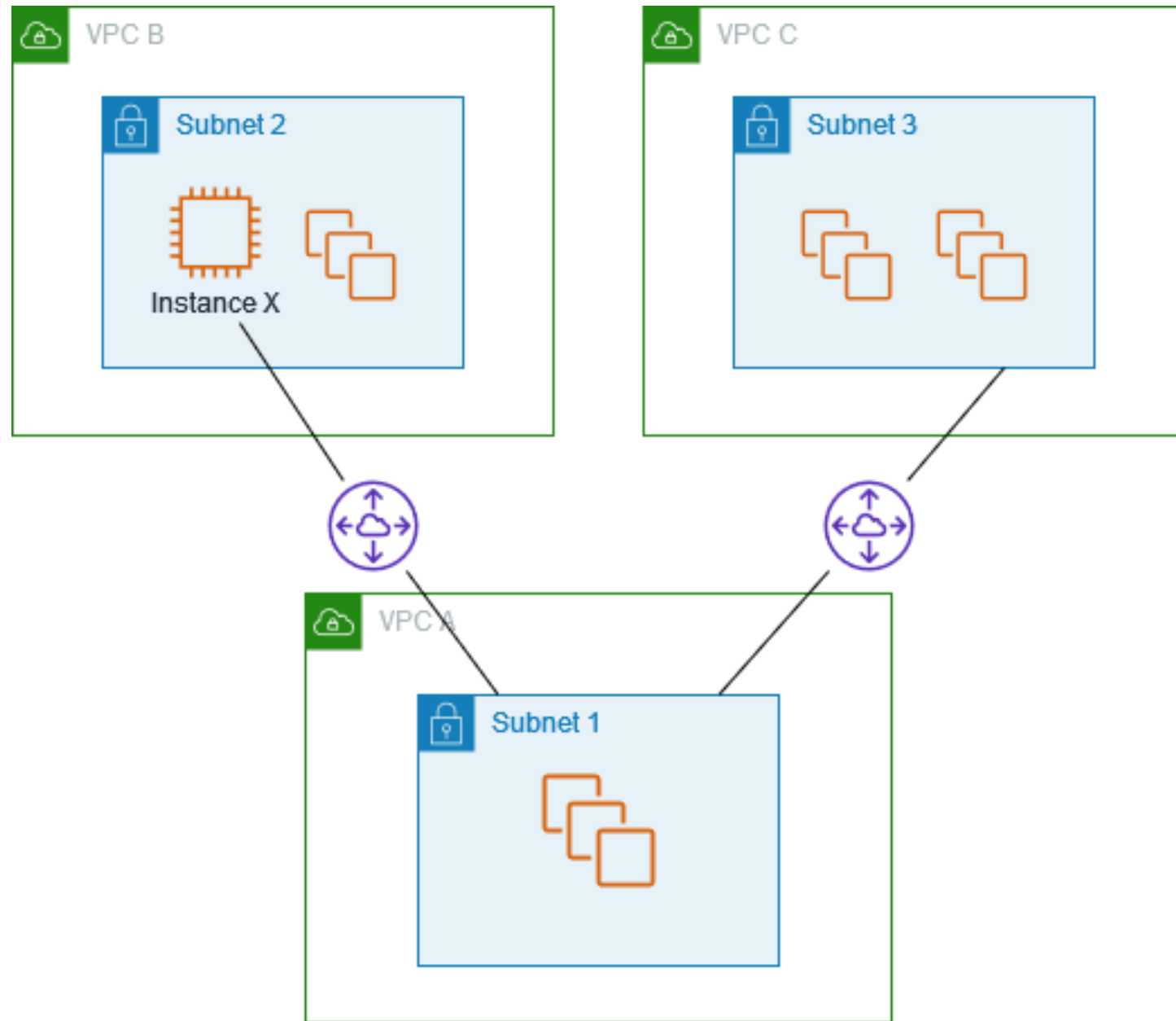
Network Transit Gateway

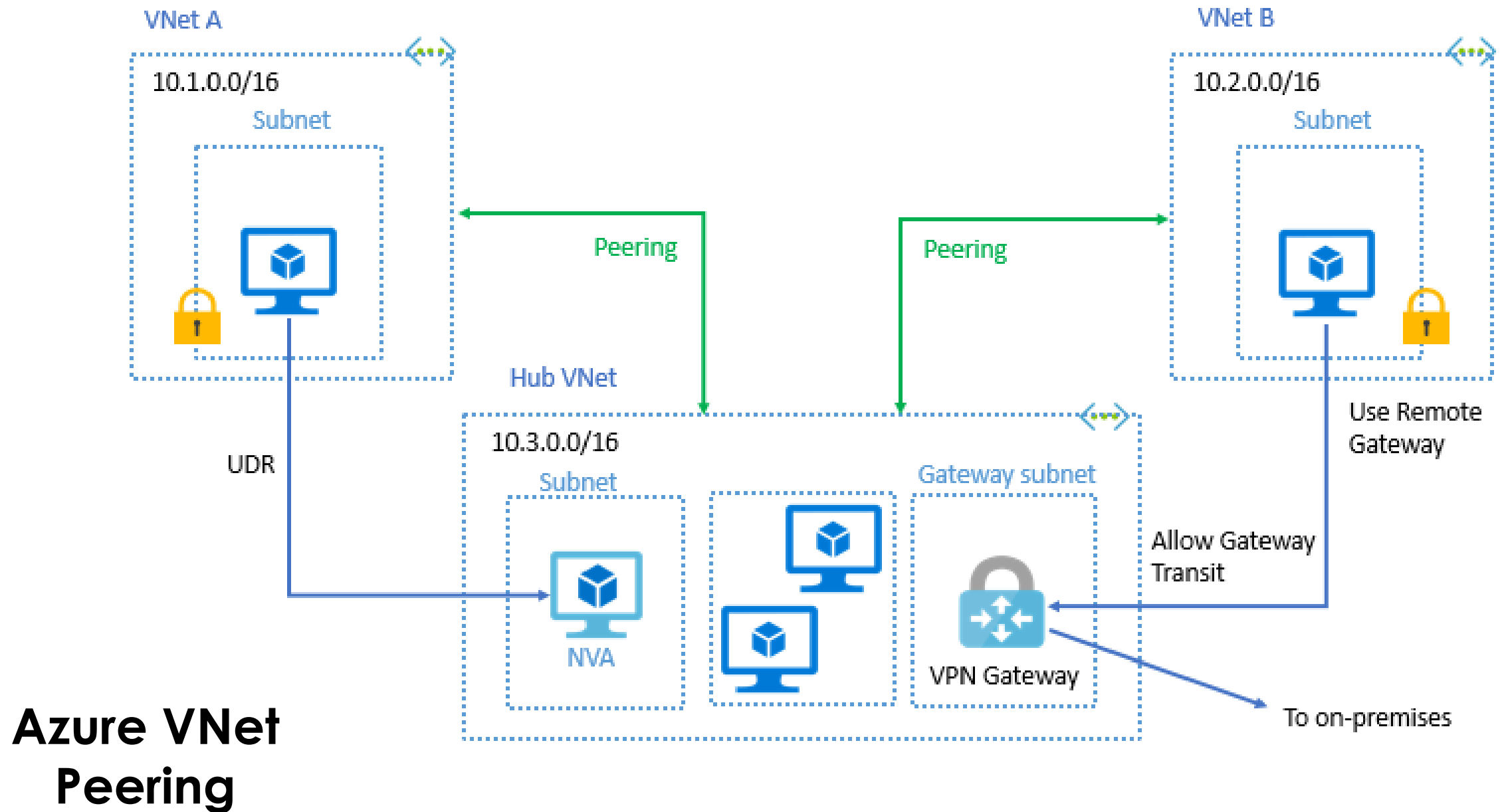
- A service which connects multiple virtual networks and on-premises networks through a central hub.
- This architecture used to be implemented through the hub-spoke model.
- Implementations
 - AWS Transit Gateway
 - Azure Virtual WAN

Network Peering

- Allows to connect and communicate between two virtual networks.
- Used to be implemented through VPN Gateways.
- Each virtual network can be peered to more than one virtual networks.
- Can peer virtual networks across accounts / subscriptions.
- Does not support transitive peering.
 - If VNet A is peered with VNet B and VNet B is peered with VNet C, VNet A cannot communicate with VNet C and vice-versa
- Both AWS and Azure have implemented this feature

AWS VPC Peering



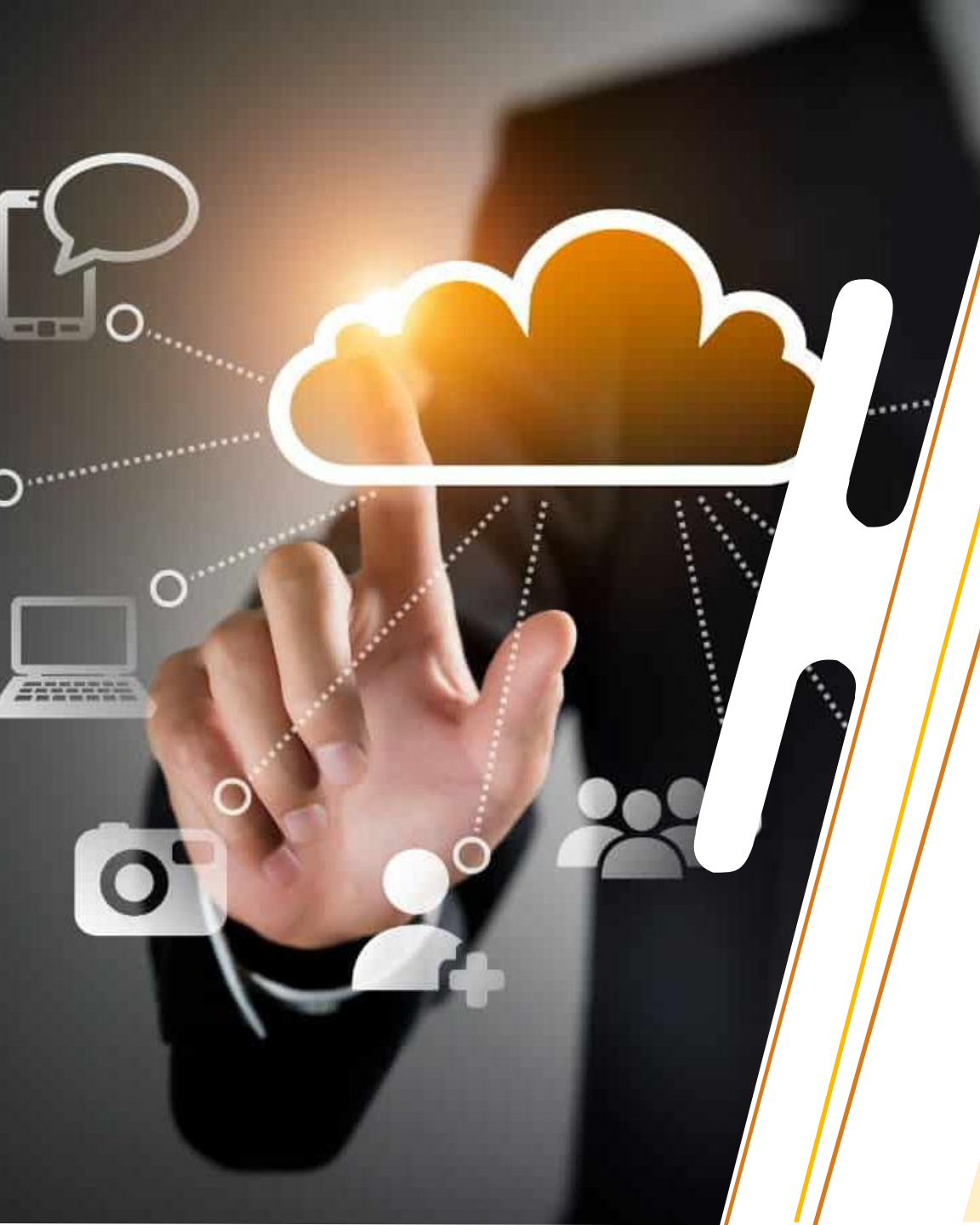




Demo

References

- AWS Academy Cloud Foundations
- AWS Documentation
- Azure Documentation



Next Session :
Cloud Storage